

Design by Falsification: Distilling an Evidence-Bound Multi-Agent Harness from the Registered Failures of a Council-of-Experts Pipeline

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Abstract

Multi-agent pipelines that route a task through domain-specialist language models and synthesize their contributions through a lead writer rest on several premises: that specialists contribute knowledge a generalist lacks, that instructions can make the writer epistemically careful, and that a model can recognize when it is out of its depth. This paper reports a falsification-first evaluation of such a council-of-experts architecture across more than fifty pre-registered experiments on auditable local hardware (7 to 20B parameter models), in which nearly every premise failed its registered test. Specialist fine-tunes carry per-model rather than per-domain signatures; an epistemic instruction installs only its named phrase; preference judges then penalize that phrase; stable model self-reports (confidence, priors, ownership) do not predict behavior; and prose transports only 0.11 to 0.33 of the epistemic qualification supplied to it. The falsifications localize mechanisms, and the mechanisms compose into an orchestration harness in which redundancy is engineered rather than assumed, epistemic qualification travels outside the writer’s prose, re-consultation is triggered by mandated artifacts rather than self-assessment, and no component is retained on rationale alone. Each load-bearing design rule was then re-tested prospectively as its own registered experiment. Assigning a contested quantity to a second specialist halves corrupted-value adoption (from 0.900 to 0.450); an appendix assembled outside the writer carries 100% of specialist caveats against the 17% that survives prose, changes downstream decisions (+0.691 over a matched irrelevant control) at parity with in-prose placement, and shows no detected preference cost; and the full re-consultation loop closes with a live specialist (0.905 adoption versus 0.333 control, within noise of a scripted ceiling). Two registered tests contradicted the design rationale: sibling isolation lost its contamination warrant to an informative null, and an elicited ownership rule failed its confirmation experiment at exactly chance. Both demotions are recorded in place. The paper argues that this acquisition path, retaining only components that survive registered attempts to falsify them, is the appropriate standard for multi-agent architecture.

1 Introduction

The architecture under study follows a design that was conventional in 2024 and 2025. Three domain-specialist models (healthcare, legal, finance) each analyze a decision problem from their own expertise, and a lead writer synthesizes their contributions. Aggregation pipelines of this shape report state-of-the-art preference results [1]; specialist fine-tunes are widely available; and the architecture’s premises appear almost too obvious to test. Specialists contribute what

generalists cannot. Instructing the lead to preserve uncertainty makes it careful. A model that is out of its depth can say so. Corroboration among specialists strengthens claims.

This paper reports what happens when those premises are tested. The research program ran as a falsification-first ledger: every experiment was pre-registered with predictions, falsification conditions, and attainability computed from the archive before any run; every deviation and verdict was recorded; and the instruments themselves were audited, producing fourteen recorded instrument-validity findings. A companion paper [2] reports the audit findings in full. The present paper makes the systems argument: it describes what remains of a multi-agent architecture after most of its premises fail registered tests, and what can be constructed from the parts that survive.

The argument proceeds in four stages.

1. **The premises failed** (Section 4). They failed registered experiments with pre-stated kill conditions, most of which fired.
2. **The failures localized mechanisms** (Section 5). Each falsification identified a real mechanism: the lead is a source selector rather than a verifier; instruction compliance is not behavior; and the scarce resource in consultation is the trigger, never the capability.
3. **The mechanisms compose into a harness** (Section 6). If protection is coverage, coverage can be engineered. If prose loses epistemic content, that content need not be shipped in prose. If self-assessment is empty, triggers can be drawn from mandated artifacts instead.
4. **The harness was then tested component by component** (Section 7). The load-bearing components survived with causal or predictive effect sizes; one design rationale was demoted by an informative null; one candidate rule was rejected by its own confirmation experiment; one gate failed its frozen criteria twice and is recorded as not deployable; the accuracy question was resolved as untestable for lack of headroom; and the assembled bundle was compared head-to-head against the original architecture and ships its machinery at no detectable preference cost. The architecture that remains is smaller than the one designed and better than the one assumed.

Methodological claim. Multi-agent frameworks commonly ship components on rationale: a verification stage because verification sounds prudent, a confidence threshold because calibration sounds measurable, a debate round because deliberation sounds epistemic. The central claim of this paper is that every component of a multi-agent system is an empirical claim about model behavior, and that most such claims are false at the scale where practitioners actually deploy them. The harness presented here is not offered as the best possible architecture. It is offered as the subset of one architecture that survived more than fifty registered attempts on its premises, with the verdicts, including the demotions, attached to the components they license.

Contributions. The paper makes the following contributions.

1. A ledger of nine architectural premises for specialist aggregation pipelines, each with its registered test and verdict, in which seven are falsified, one is null, and one is resolved as untestable at the studied scale (Section 4).

2. An orchestration harness in which each component is derived from a specific verdict and does only what that verdict licenses (Section 6).
3. Prospective single-factor re-tests of each load-bearing harness component, with effect sizes, confidence intervals, and two recorded demotions of the design’s own rationales (Section 7).
4. A head-to-head comparison of the assembled harness against the original architecture under a validated order-debiased two-family preference protocol (Section 7).
5. A statement of operating economics and open risks, each attached to the component it concerns (Section 8).

2 Related Work

Aggregation pipelines. Mixture-of-Agents [1] and its successors report preference gains from aggregating multiple proposer outputs through an aggregator model. Self-MoA [3] shows that mixing weaker models often degrades results, a finding consistent with the per-model signature result reported here. Debate and multi-agent discussion studies [7] find gains concentrated where prompts were weak, consistent with the finding that the one intervention retained here routes information rather than deliberation.

Judge pathologies. Critiques of LLM-as-a-judge evaluation [4] motivate the order-debiased two-family protocol used throughout. Length and sycophancy pressures under preference optimization [5, 6] sharpen into the measured phrase penalty reported in the companion audit.

Pre-registration in systems research. No prior work is known to the author that pre-registers each component of a multi-agent architecture and reports the failures alongside the survivals. That practice, more than any single component, is what this paper proposes for adoption.

3 Background

3.1 The council-of-experts pipeline

The audited pipeline is a two-layer aggregation architecture. Three specialist seats, each a prompted or domain-fine-tuned 7 to 8B model or a role-prompted sample of the writer, receive a decision problem and produce independent contributions. The contributions are concatenated and passed to a 20B mixture-of-experts writer (`gpt-oss:20b`) operating as the lead, under either a published aggregate-and-synthesize prompt or a minimal lead-analyst prompt. The case battery consists of eighteen strategic advisory scenarios spanning healthcare, legal, and finance content.

3.2 Registration protocol

Every experiment in the program is registered before it runs. A registration states the hypotheses, the falsification conditions, the attainability computation (the fraction of runs in which the outcome variable can fire, computed from existing data for every registered prediction), and the consequences of each verdict. Verdicts are entered in a status ledger that distinguishes live,

provisional, withdrawn, and blocked claims. Three program-level audits re-classified earlier results, and null results are cited only at the strength their audit class permits (informative, weak, uninformative, or degenerate). Registrations are committed to the repository before first runs, so temporal ordering is verifiable from version history.

3.3 Terminology

The ledger refers to each registered experiment as a *cell*; the term is retained here where a specific ledger entry is cited (for example, C55). A *seat* is one specialist position in the pipeline. The *lead* is the synthesizing writer. The *orchestrator* is the non-model control layer that plans, dispatches, and assembles. *Transport* denotes the fraction of epistemic qualification supplied upstream that appears in the lead’s output. Confidence intervals are cluster-bootstrap intervals over cases unless stated otherwise.

4 Premises That Failed Registered Tests

Table 1 summarizes the program’s negative results. Each row lists a premise the council architecture requires, the registered experiment that tested it, and the verdict. Effect sizes and protocols are reported in the companion paper; here each premise is given one sentence of mechanism.

Three of these results warrant additional detail, because the harness is built directly on them.

Compliance is not behavior. The phrase-swap design placed two synonymous instruction forms against a control whose bytes differ only in the instructed clause. The instruction reliably produced its own phrase and nothing measurable beyond it. A dictated-phrase registry with a measured zero over-attribution rate (0/2,907 spans) guarantees the accounting. On this evidence, every instruction of the form “be careful” that a framework ships is purchasing surface forms.

The lead is a source selector. With a clean co-source covering a contested quantity, corruption never propagated (0/27); with no clean source, it propagated or the quantity was silently dropped. The lead does not check; it chooses. This single mechanism sets both the ceiling (no verification will emerge from prompting) and the opportunity (choice can be engineered by controlling which sources exist).

The trigger is the scarce resource. In every consultation design measured, the mechanism worked when fired and the firing was rare. The lead uses a routed clarification in 7/7 runs but names the tension that warrants it at 0.34; specialists route correctly at 0.909 but recognize the need at 0.306; across three experiments the trigger rate ranged from 0.208 to 0.583 by batch. Self-assessment, the faculty that frameworks rely on for escalation, was the one faculty the program never found.

5 Effects That Survived

Three measured gains outlived the audit, and they are the only three the harness claims.

Aggregation wins preference. The pipeline beats the same writer answering directly in 0.818 [0.718, 0.919] of decisive order-debiased pairs, surviving length matching and replicating under a disjoint judge family (0.788). The composition of this lift is not explained by the full surface-feature set and is reported as unidentified. The lift is nonetheless real, and it is not attributable to the epistemic behaviors the prompts request.

Error immunity where coverage overlaps. The source-selector mechanism, read constructively: 0/27 propagation when an uncorrupted co-source exists.

Routed re-consultation resolves contested quantities. When an orchestrator dispatches a follow-up and the reply is informative, the lead adopts the implied resolution at 1.000 (scripted) and 0.905 (live specialist) against 0.333 without the loop. A length-matched contentless reply lands at 0.200, so the information does the work rather than the ritual.

6 The Harness

Each component described below is derived from a specific verdict in Section 4, cites it, and does only what that verdict licenses. Figure 1 shows the assembled pipeline with every edge labeled by its measured value. The full architecture document ships with the ledger.

6.1 Design principles

The harness follows four principles, each licensed by a verdict.

1. **Engineer coverage rather than assume verification.** The source-selector result establishes that protection against error is a function of which sources exist, not of any checking the lead performs.
2. **Carry epistemic content outside the writer.** The transport bound establishes that prose loses two thirds to nine tenths of the qualification supplied to it, and the preference result establishes that in-prose marking is penalized.
3. **Trigger from artifacts, not self-assessment.** The consultation results establish that models act correctly when dispatched but do not reliably recognize the need to be dispatched.
4. **Retain nothing on rationale alone.** Every component carries the registered verdict that licenses it, including verdicts that demoted the original design rationale.

6.2 Components

Seating is empirical, never categorical. Because signatures are per-model, no seat is chaired for its category label. A candidate passes a measured gate consisting of a degeneracy screen (at least 800 visible characters on screen items; one archive fine-tune produced 12/12 degenerate probe runs), a tuned-minus-own-base signature probe, and a format smoke test at production token budgets. A generalist under a role prompt is a legitimate seat if it passes; most experiments in the program ran on exactly that configuration.

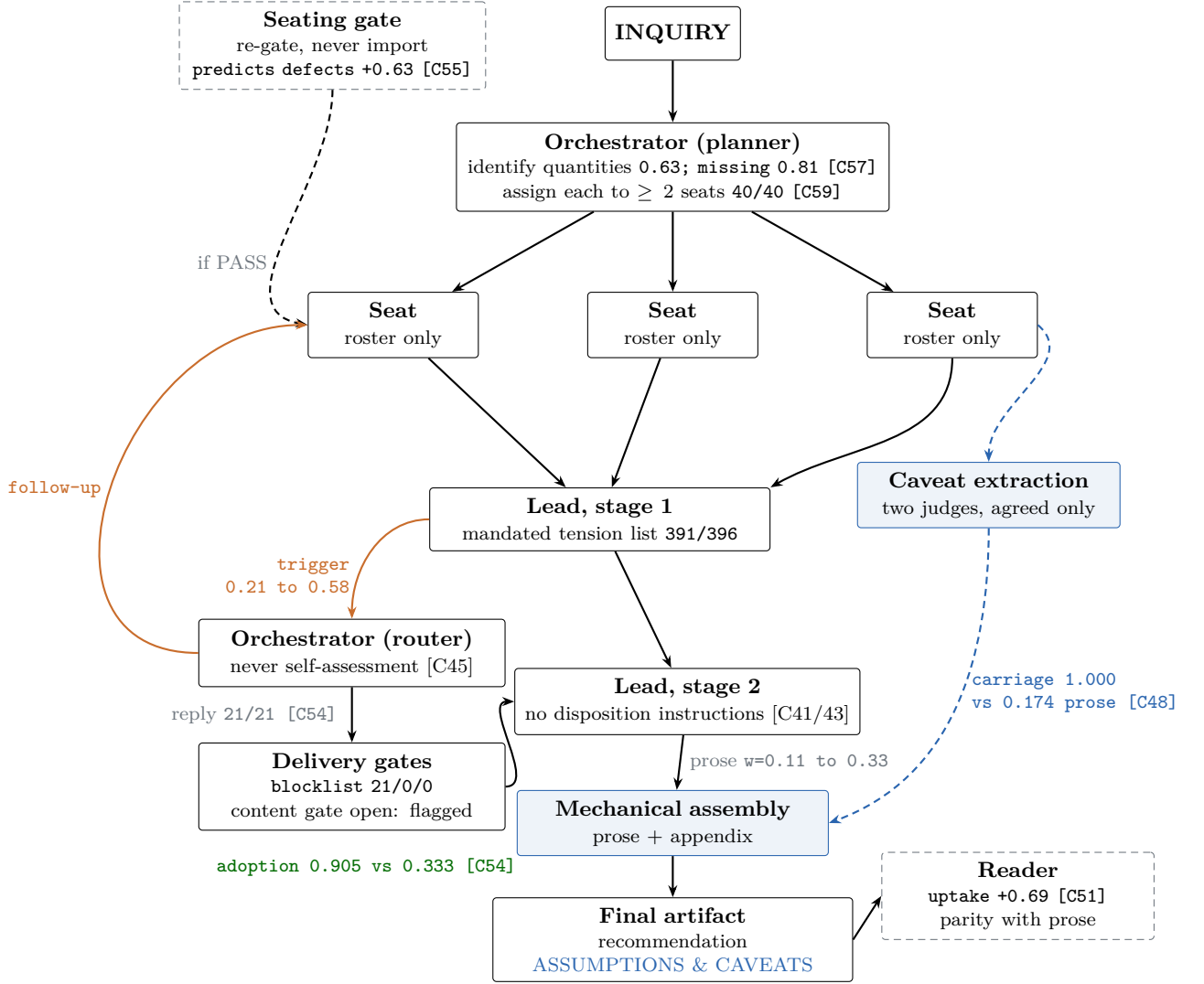


Figure 1: The assembled harness on one inquiry, with every edge labeled by its registered measurement. Dashed blue: the out-of-band channel for epistemic qualification, which the writer never sees. Orange: the orchestrator-routed re-consultation loop, bounded by its measured trigger rate. Seats are interchangeable by industry; chairs are earned by the gate, never by category label. Engineered overlap between seats is where the error immunity resides: corrupted-value adoption falls from 0.90 to 0.45 [C47], and the clean co-source is used end-to-end at +0.43 [C59].

Isolation, on cost grounds only. Seats receive their task and a one-line roster. Sibling outputs are not forwarded. The original contamination rationale for this rule was withdrawn (Section 7); the rule survives because sibling context costs approximately 10k input characters and 14% longer outputs per seat while purchasing no measured change in lane discipline.

Redundancy is engineered. The orchestrator identifies load-bearing quantities in the decomposition and assigns each to at least two seats. This is where the council’s error immunity resides; it does not exist by default.

Triggers come from mandated artifacts. The lead runs in two stages: a mandated tension list first (391/396 archived runs produce one when required), then synthesis. The orchestrator, never any model’s disposition or self-assessment, parses the list and dispatches follow-ups. A reply that adds nothing beyond the seat’s first contribution is flagged rather than delivered silently (a ritual-effect risk observed at $d \approx 0.22$, below the power to confirm, is treated as live).

Epistemic qualification travels out-of-band. Caveats, assumptions, and confidence never route through the writer. Seats emit them as structured fields, and the harness carries them mechanically to an appendix of the final artifact. The writer receives no disposition instructions at all: the evidence indicates that such instructions purchase nothing and that their surface forms cost preference.

A Goodhart charter. Trigger rates, hedge counts, compliance shares, transport slopes, and preference scores on epistemic content are diagnostics and may never be optimization targets; the optimizable signal must live outside the pipeline’s own text. Each entry on the forbidden list cites the experiment that showed what optimizing it would manufacture.

7 Prospective Re-tests of the Harness

Design rules distilled from findings can still be wrong, because post-hoc patterns calcify. Each load-bearing rule therefore received its own registered experiment, with kill conditions stated in advance. The verdicts follow.

7.1 The seating gate predicts pipeline defects (survived)

Ten local candidates were gated on the frozen degeneracy screen, using two screen items disjoint from the pipeline case set, with verdicts committed to the repository before any pipeline run. Every candidate was then seated on the pipeline cases. Strict rank separation held: both gate-fail candidates defected in the pipeline (0.333 and 0.944 of contributions) while all eight gate-pass candidates ran at 0.000 to 0.056; the pooled contrast is +0.632 [+0.549, +0.722]. Two refinements followed. First, gate verdicts age: one model that passed the archive-era screen failed the fresh re-gate and then defected in-pipeline at 0.333, so seating must re-gate rather than import a stored verdict. Second, format compliance is a separate axis that does not predict seat defects; the legal fine-tune failed the format smoke test 0/4 yet seated defect-free, so that component’s proper target is verdict-emitting roles.

7.2 The planner chain works, measured link by link (survived, with bounds)

The redundancy planner was the one component that had never been exercised: every prior redundancy result used human-planted assignments. Three experiments closed the gap. *Identification*: the orchestrator recalls load-bearing quantities at 0.633 [0.506, 0.758] against analyst-authored labels (a thirty-fold specificity separation over a mismatched-case control), unchanged by seeing the specialists' contributions, so it runs at decomposition time. Its best class is the right one: *missing* quantities (unmeasured rates, untracked inputs) are identified at 0.813 against 0.531 for stated figures. *Assignment and use, fully live*: given the identified quantity, the planner writes it into at least two sub-questions in 40/40 plans; live seats convey briefed values at 0.85 to 0.95; and the writer uses the clean co-source end-to-end (+0.425 [+0.325, +0.525] clean adoption against the planted +0.500). A halted first attempt produced a channel finding: privately briefed facts convey only into task-relevant assignments (2/8 when orthogonal against 21/21 when pointed), a constraint the corrected planner input satisfies by construction. Expected end-to-end protection is therefore the product of identification coverage (approximately 0.63) and the selection effect.

7.3 The content gate failed (recorded as not deployable)

The rule that follow-up replies adding nothing should be dropped failed two registered attempts on complementary halves. A two-judge semantic gate passes vacuous fillers (3/6 dropped), and an extract-then-verify gate that fixes vacuity (6/6) rejects 52% of genuine live replies. Under the frozen commitment there is no third attempt without purpose-built labeled data; the interim posture is flagged delivery, never silent gating. The fabrication blocklist gate, by contrast, is deployable: 21 true hits, 0 false positives, and 0 misses over the 141-reply archive, seeded only from committed classifications. The failed gate is reported beside the deployed one because refusing that asymmetry is the method.

7.4 Redundancy is causal (survived)

A single factor, assigning the contested quantity to a second seat, halved corrupted-value adoption from 0.900 to 0.450 ([+0.200, +0.725]), with the clean value actually adopted (from 0.000 to 0.500). This is selection rather than dilution. The bare arm's 0.900 also stands as the sharpest no-verification measurement on record. Protection is bimodal across items; two candidate boundary mechanisms were then registered and rejected (prior plausibility at 0.544; the elicited ownership map at exactly 0.500 prospective against 0.895 descriptive, with a minimum detectable difference of 0.68, the confirmation experiment performing its anti-calcification function). The boundary is open and stated as such.

7.5 Out-of-band qualification carries and is used (survived)

Carriage: the appendix delivers 1.000 of both-judge caveat sentences against 0.174 surviving prose, with no detected preference penalty (0.510 [0.267, 0.754]; not evaluable at power, reported as bounded). *Uptake*: appendix-delivered caveats flip a reader's registered decision at 0.727 against 0.036 for the identical block carrying an irrelevant caveat, with direction-balanced items so that caution cannot masquerade as uptake, at parity with in-prose placement (Figure 2). The channel that preference judging cannot see is not an exhibit; it changes decisions.

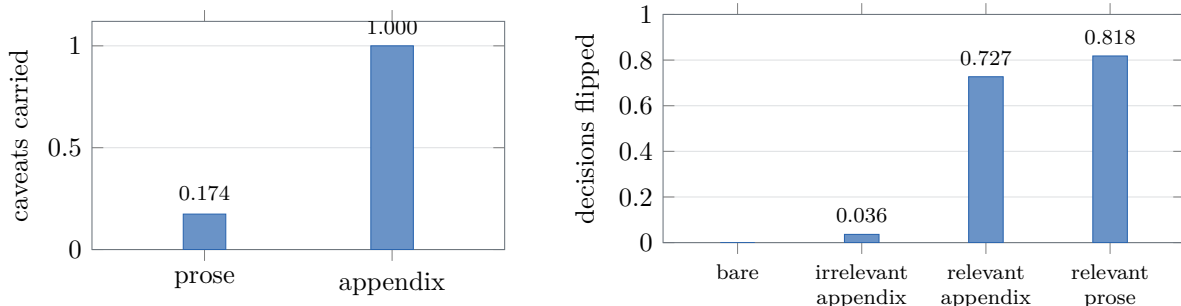


Figure 2: The out-of-band channel, tested on both halves. *Left*: caveat carriage through writer prose against the mechanically assembled appendix (zero-generation design). *Right*: registered two-token reader decisions flipped by a planted decision-relevant caveat, by delivery arm; the irrelevant-appendix control shows that the effect is content rather than appendix presence. Uptake is $+0.691$ [$+0.455$, $+0.891$]; parity with prose is $+0.091$ [-0.036 , $+0.218$].

7.6 The live loop closes (survived, with one halt)

A first live-specialist experiment buried the deciding fact in the specialist’s own round-one text and halted at its registered pilot gate: the trigger named the tension at only 0.208 with the fact visible in the pile. The redesign supplied the fact as the specialist’s private working notes, knowledge held but never written, and restored the original round-one bytes. The live specialist conveyed the fact 21/21 times when dispatched; the lead adopted at 0.905 against 0.333 archived control ($+0.571$ [$+0.139$, $+0.912$]), within -0.095 [-0.227 , 0.000] of the scripted ceiling. One new risk is on record: 4/21 live replies embellished the correctly conveyed fact with fabricated authority (invented case law, invented regulatory rulings) and were adopted regardless. Content gates screen absence, not invention; the fabrication rate was subsequently bounded (Section 8).

7.7 Isolation lost its warrant (demoted)

The registered one-factor test made byte-fixed sibling contributions visible under production prompts. Off-domain framework density moved -0.009 [-0.048 , $+0.028$] against a pre-registered minimum detectable difference of $+0.05$ to $+0.08$ computed from the archive, an informative null, with in-domain density unchanged. The contamination rationale is withdrawn; the rule stands on cost. A registered secondary measure pointed in the opposite direction: seats shown their siblings’ coverage used slightly fewer off-domain terms of their own (-0.033 [-0.053 , -0.013]). This is recorded as a candidate mechanism, not claimed.

7.8 The ownership rule failed confirmation (rejected)

A stable, near-unanimous elicited map of which seat “owns” each quantity, exactly the kind of pattern a framework would ship as a routing table, predicted arbitration at chance on fresh items. This rejection is reported with the survivals because it is the argument: the difference between an evidence-bound harness and a prompt stack is that the harness can lose these arguments on the record, component by component.

7.9 The assembled bundle shows no detectable preference cost (survived)

With every component separately verdicted, the remaining deployment question was head-to-head: the assembled harness’s artifact against the original architecture’s, on the validated order-debiased two-family protocol (108 pairs). The result is no evaluable difference: primary-judge share 0.562 [0.375, 0.765], replication 0.333 [0.083, 0.562], both spanning 0.5, with large effects excluded in both directions. The harness’s preference standing is inherited from aggregation itself, which both bundles share; its differentiators were never preference plays, and shipping them (five caveats per artifact, a follow-up round, restructured seats) costs nothing detectable. The appendix’s within-pair attribution is mildly positive (+0.073 toward the harness). One descriptive statistic from the same experiment reframes the loop’s economics: on organic inquiries the orchestrator dispatched a follow-up on 46/54 pipelines (0.85). Real disagreement between seats is abundant, against the 0.21 to 0.58 trigger range measured on planted tensions.

8 Operating Economics and Open Risks

The trigger bounds coverage, less than anticipated. Everything downstream of the trigger is measured and works. On planted tensions the trigger fires at only 0.208 to 0.583 by batch, and it is deliberately not an optimization target, because raising it blindly manufactures the ritual condition that the loop’s own control arm exposed. The bundle comparison’s telemetry reframes the economics: on organic inquiries, where seats genuinely disagree, dispatch fired on 0.85 of pipelines. The scarcity that bounded the experimental cells largely dissolves on natural problems; planted-tension rates are the floor rather than the forecast.

Open risks. Per-item protection is bimodal with the boundary mechanism unknown (two candidates rejected). Live-reply fabrication is bounded by a registered follow-up experiment: reply-level 0.100 [0.047, 0.201] with the deciding fact in hand and 0.150 [0.081, 0.261] without it, concentrated entirely on one trigger topic (15/20 there, 0/100 elsewhere) and name-stable, with the same two invented decisions recurring across independent samples, which licenses a blocklist delivery filter. The seat never admits the gap (0/20 sampled), the backfill mechanism is directionally positive but not evaluable at six clusters, and the bound covers adversarial case citations only, a floor rather than a ceiling on confabulation. The preference cost of the appendix is bounded but not evaluable at current power. The council’s accuracy question is resolved as untestable for lack of headroom: two registered batteries (multi-step computation; rule-path selection) placed the direct writer at ceiling (0.977, 0.983) under baseline-only difficulty screens, so no well-specified known-answer battery can measure a council difference at this scale. This is the strongest available statement of the design rule that a council is not an accuracy device. None of these risks is hidden in the architecture document; each is attached to the component it concerns.

Running the harness replicates the science. The harness’s telemetry doubles as a replication battery. Every deployment yields transport estimates, trigger telemetry, phrase-swap slots for proposed instructions, and preference-versus-verified divergence wherever ground truth exists, each with its attainability gate. A faithful deployment is also, at near-zero marginal cost, the external validation that the program’s scope caveats call for.

9 Limitations

The scope is 7 to 20B local models, advisory-domain English, and one primary writer family, with cross-family replications where stated. Cluster ceilings of 6 to 18 items bind every harness experiment; the live-loop comparison uses archived arms under a registered fresh-batch check; the uptake verdict covers 11 of 18 authored items after a registered floor gate; and trigger rates are batch-variable and quoted only as a range. The harness’s gains are preference, immunity, and resolution, not accuracy, which two registered batteries resolved as untestable at this scale: the direct writer ceilings at 0.977 to 0.983 on well-specified items, so no headroom exists (under-specified or adversarial constructions would pose a different question). Verdicts inherit their experiments’ scope and should not be generalized past it; the architecture document states, for every component, exactly which registered experiment licenses it.

10 Conclusion

A council-of-experts pipeline was built on premises that nearly all failed registered tests. The failures were informative: each localized a mechanism, and the mechanisms composed into a harness whose every component is licensed by a specific verdict. Prospective re-tests confirmed the load-bearing components with causal or predictive effect sizes, demoted one design rationale, rejected one candidate rule, recorded one gate as not deployable, and found that the assembled bundle ships its machinery at no detectable preference cost. The resulting architecture is smaller than the one designed and better supported than the one assumed. The practice proposed for adoption is the acquisition path itself: retaining only what survives registered attempts to falsify it, and reporting the demotions in place beside the survivals.

Reproducibility Statement

The complete pre-registration ledger (52 registration blocks at this writing), every deviation and verdict including the two halts, the architecture document, the measurement kit (`gst`), the frozen dictation registry, and all item files are in the project repository. Registrations are committed before first runs, so temporal ordering is verifiable from version history.

References

- [1] J. Wang, J. Wang, B. Athiwaratkun, C. Zhang, J. Zou. Mixture-of-Agents enhances large language model capabilities. *arXiv:2406.04692*, 2024.
- [2] S. Bobo. Instructions buy the phrase; the phrase costs preference: an epistemic audit of LLM aggregation pipelines at auditable scale. Companion draft, 2026.
- [3] W. Li et al. Rethinking Mixture-of-Agents: is mixing different large language models beneficial? *arXiv:2502.00674*, 2025.
- [4] L. Zheng et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *NeurIPS Datasets and Benchmarks*, 2023.
- [5] P. Singhal et al. A long way to go: investigating length correlations in RLHF. *arXiv:2310.03716*, 2023.

- [6] M. Sharma et al. Towards understanding sycophancy in language models. *arXiv:2310.13548*, 2023.
- [7] Q. Wang et al. Rethinking the bounds of LLM reasoning: are multi-agent discussions the key? *arXiv:2402.18272*, 2024.

Premise required by the council	Test	Verdict
Specialist fine-tunes share a domain signature	two medical tunes of one base versus their base	falsified: signatures are per-model; the tunes disagree with each other
An epistemic instruction changes the lead’s behavior	counterbalanced phrase-swap; synonym forms	falsified: the named phrase appears (0.67 to 0.70); the construct beyond it is unchanged (CIs $[-0.079, +0.111]$, $[-0.103, +0.087]$)
Careful marking is at worst neutral under evaluation	order-debiased preference, phrase arms versus own control	reversed: the phrase <i>loses</i> preference (0.261, 0.317; replicated cross-family)
A specialist can flag its own limits	roster plus deferral token; in-domain versus out-of-domain	falsified: routing is accurate when fired (0.909) but discrimination spans zero (0.306 versus 0.083)
Corroboration among specialists strengthens transport	planted agreement across contributions	null: 0.487 versus 0.524, CI spans zero
The lead verifies or recomputes	corrupted-source injection	falsified: the lead <i>selects</i> sources; 0/27 propagation with a clean co-source, 5/27 when all sources are corrupt
The council improves accuracy	three verifiable-outcome batteries	untestable: the direct writer is at ceiling (0.92 to 0.98, including registered computation and rule-interaction rounds under baseline-only difficulty screens); no headroom exists for a difference
Prose carries the specialists’ epistemic content	sentence-level transport, two writers, two corpora	bounded: slopes $w = 0.11$ to 0.33; two thirds to nine tenths of qualification is lost in synthesis
Elicited model self-state predicts behavior	priors (0.544); elicited ownership map	falsified twice: the ownership map was stable and near-unanimous, and predicted arbitration at exactly 0.500 against a descriptive 0.895

Table 1: The premises, their registered tests, and their verdicts. Full protocols, confidence intervals, and pre-registrations are given in the companion paper and the public ledger.